

AI-POWERED HR ASSISTANT

NLP-Based Employee Query Classification & Response System

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PROJECT REPORT

Artificial Intelligence • Natural Language Processing • Human Resources

1. Executive Summary

SmarterHR AI is an NLP-based HR assistant designed to recognize common employee questions and return predefined HR guidance. The system uses TF-IDF text features, Logistic Regression intent classification, confidence-based fallback handling, and a structured HR response knowledge base. A Streamlit interface provides a practical browser-based experience.

The system contains 107 training examples across 16 HR intents. The recorded hold-out evaluation uses an 80/20 train-test split and reports 63.64% accuracy on 22 test examples. Because the dataset is relatively small, these results represent a prototype baseline and should be validated with a larger, more representative dataset before production use.

107	16	80 / 20	63.64%
Training examples	HR intents	Train / test split	Recorded accuracy

Project objectives

- Automate repetitive HR questions and provide fast first-line guidance.
- Use NLP-based intent classification instead of keyword-only rules.
- Separate intent detection from response generation through a knowledge-base dictionary.
- Package the trained components into a reusable model artifact.
- Provide a professional Streamlit interface suitable for internal deployment.

2. Solution Architecture

The system follows a clear pipeline from an employee's natural-language question to an HR response.

Employee Question Natural-language query	TF-IDF Text → features	Logistic Regression Intent prediction	Confidence Check Threshold = 0.35	Knowledge Base Intent → response
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Core technologies

Component	Technology	Role
Data handling	Pandas	Creates and manages the labeled text dataset.
Feature extraction	TfidfVectorizer	Transforms text into TF-IDF features using unigrams and bigrams.
Classifier	Logistic Regression	Predicts one of the defined HR intents.
Workflow	scikit-learn Pipeline	Combines vectorization and classification.
Persistence	Joblib	Stores the model package and chatbot configuration.
Interface	Streamlit	Provides the browser-based HR assistant interface.

3. Dataset & Intent Coverage

The system uses 107 labeled employee questions covering 16 HR intent categories. The distribution is shown below.



Intent categories

Intent	Examples
attendance	9
leave	9
salary	8
benefits	7
recruitment	7
interview	7
onboarding	6
payroll	6
resignation	6
work_from_home	6
company_policy	6
hr_contact	6
experience_letter	6
greeting	5
thanks	5

goodbye	4
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4. Machine Learning Methodology

4.1 Text representation — TF-IDF

TfidfVectorizer uses unigrams and bigrams with sublinear term frequency scaling. This converts employee questions into numerical features while retaining useful phrase-level information.

4.2 Classification — Logistic Regression

Logistic Regression is configured with $C=5$ and $\text{max_iter}=2000$. The classifier is combined with the vectorizer in a single scikit-learn Pipeline so preprocessing and prediction remain consistent.

4.3 Train / test evaluation

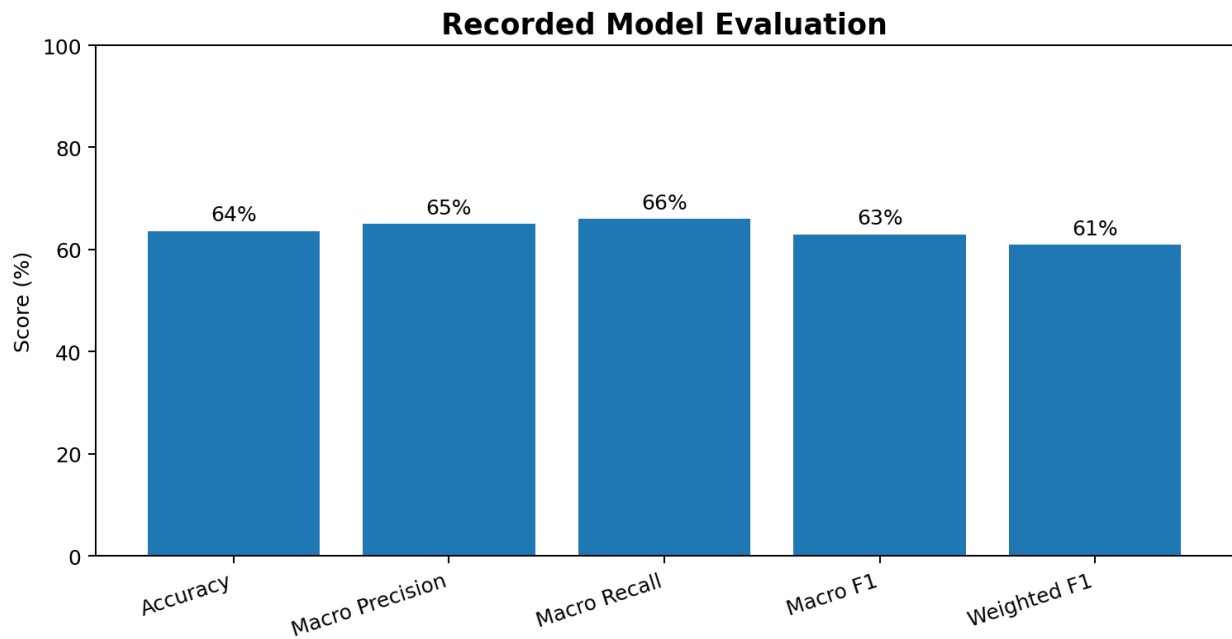
An 80% training and 20% testing split is used. The recorded test set contains 22 examples.

4.4 Confidence safeguard

A confidence threshold of 0.35 is applied. When the highest predicted probability is below the threshold, the assistant returns a fallback response instead of presenting a low-confidence classification as certain.

5. Model Evaluation

The evaluation below summarizes the recorded prototype performance.



Metric	Result	Interpretation
Accuracy	63.64%	14 of 22 test examples classified correctly.
Macro precision	65%	Average precision across the 16 intent classes.
Macro recall	66%	Average recall across the 16 intent classes.
Macro F1	63%	Balanced precision/recall summary.
Weighted F1	61%	F1 weighted by class support.

Evaluation note: several test classes have very small support, so per-class metrics are sensitive to the particular split. The results should be treated as a baseline for further development.

6. Detailed Classification Results

Intent	Precision	Recall	F1	Support
attendance	1.00	1.00	1.00	2
benefits	0.50	1.00	0.67	1
company_policy	1.00	1.00	1.00	1
experience_letter	1.00	1.00	1.00	1
goodbye	0.00	0.00	0.00	1
greeting	0.67	1.00	0.80	2
hr_contact	1.00	1.00	1.00	1
interview	1.00	1.00	1.00	1
leave	1.00	0.50	0.67	2
onboarding	0.00	0.00	0.00	1
payroll	0.00	0.00	0.00	1
recruitment	0.20	0.50	0.29	2
resignation	0.00	0.00	0.00	2
salary	1.00	0.50	0.67	2
thanks	1.00	1.00	1.00	1
work_from_home	1.00	1.00	1.00	1

The detailed report highlights strong classification for several intents on this split while some low-support classes were not correctly recognized. Increasing training examples per intent is the most direct way to improve robustness.

7. Response Knowledge Base & Application

The chatbot separates intent prediction from answer generation. Once an intent is identified, the application retrieves a predefined HR response from a structured knowledge base. This makes the response layer easy to review and maintain.

Supported HR areas

Leave	Attendance	Work From Home	Salary
Payroll	Benefits	Recruitment	Interview
Onboarding	Resignation	Experience Letter	HR Contact
Company Policy	Greetings	Thanks	Goodbye

Deployment components

Artifact	Purpose
hr_chatbot.pkl	Serialized model package containing the pipeline, responses, confidence threshold, and chatbot name.
app.py	Streamlit interface for the HR assistant.
requirements.txt	Environment dependencies for deployment.

8. Limitations & Recommended Improvements

Area	Observation
Dataset size	107 examples are modest for 16 intents.
Class imbalance	Intent counts vary across categories.
Evaluation size	The hold-out set contains only 22 examples.
Response coverage	Answers are predefined rather than dynamically retrieved from HR policy documents.
HR policy risk	Organization-specific policy validation is required before operational use.

Recommended next steps

- Expand each intent with diverse real-world phrasings and substantially more examples.
- Use a dedicated validation set and repeated stratified evaluation.
- Add confusion matrices and error analysis to guide data improvements.
- Review HR responses with qualified HR stakeholders.
- Add controlled retrieval from approved HR policy documents.
- Log anonymized low-confidence cases for continuous improvement.
- Add authentication, privacy controls, and human HR escalation.

9. Conclusion

The AI-Powered HR Assistant provides an end-to-end prototype for employee query classification, combining natural-language processing, machine learning, confidence handling, and a Streamlit user interface. Its modular architecture provides a practical foundation for future enhancement.

The recorded 63.64% accuracy establishes a measurable baseline. Future work should focus on expanding the dataset, improving evaluation reliability, validating HR responses, and introducing organization-specific policy retrieval and governance.

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